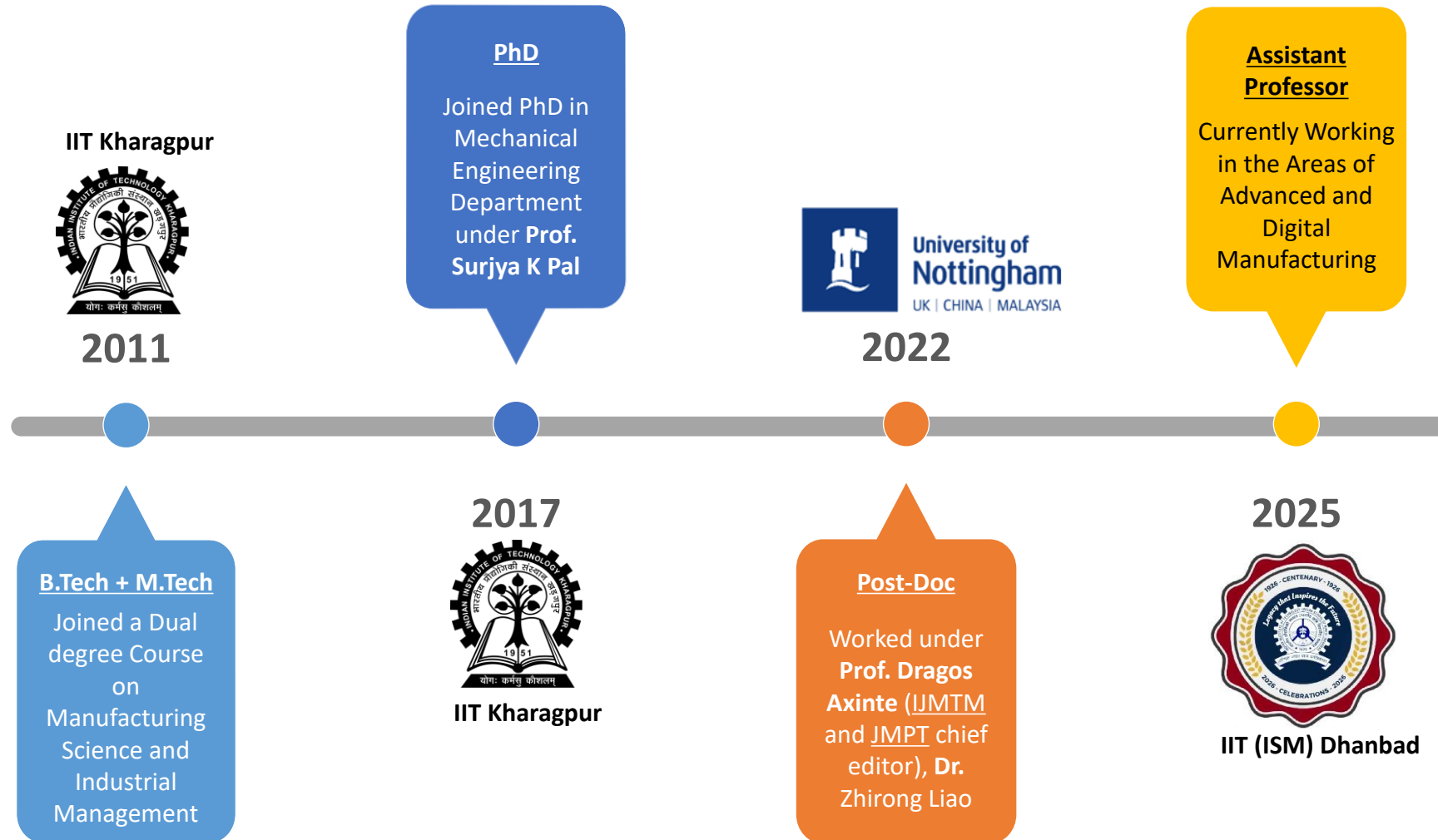


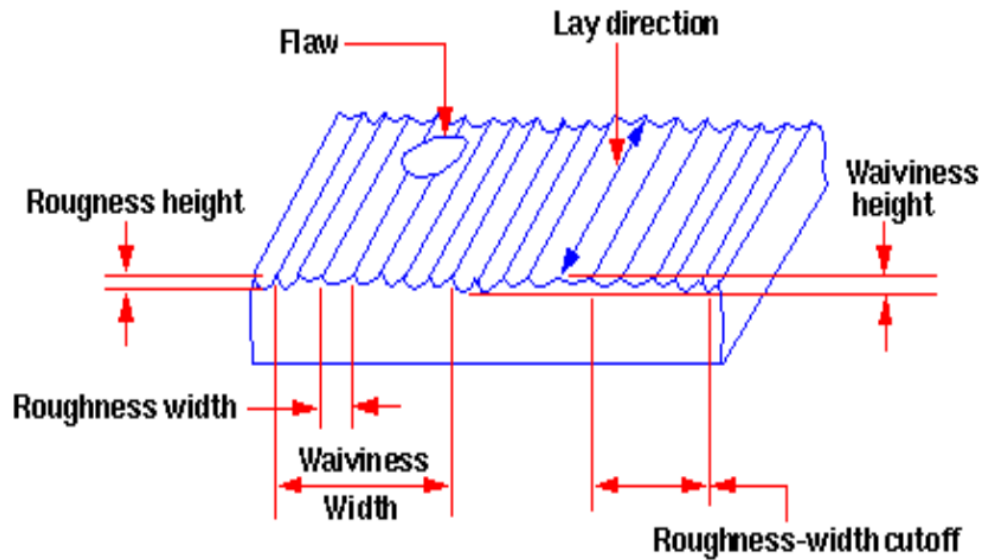
CMM and Surface Roughness Measurement

By
Omkar Mypati

About me



Terms used in surface roughness measurements



Surface characteristics (Courtesy, ANSI B46.1 - 1962)

Surface Texture

- Repetitive or random deviations from the normal surface which form the pattern of the surface. Surface texture include roughness, waviness, lay and flows.
- Every part's surface is made up of texture and roughness which varies due to manufacturing techniques and the part structure itself.
- Primary texture: $l_r/h_r < 50$ (Roughness)
- Secondary texture: $l_w/h_w > 50$ (Waviness)

1. Roughness :

Roughness consists of **surface irregularities** which result from the **various machining process**. These irregularities combine to form surface texture.

2. Roughness Height :

It is the **height of the irregularities** with respect to a **reference line**. It is measured in millimeters or microns or micro inches. It is also known as the height of unevenness.

3. Roughness Width :

The roughness width is the **distance** parallel to the nominal surface **between successive peaks** or ridges which constitute the predominate pattern of the roughness. It is measured in millimeters.

4. Roughness Width Cut Off :

Roughness width cut off is the **greatest spacing** of respective surface irregularities to be **included in the measurement** of the average roughness height. It should always be **greater than the roughness width** in order to obtain the total roughness height rating.

5. Lay :

Lay represents the **direction of predominant surface pattern** produced and it reflects the machining operation used to produce it.

6. Waviness Height:

Waviness height is the **peak to valley distance** of the surface profile, measured in millimeters.

7. Arithmetic Average (AA):

If X is the measured value from the profilometer, then the AA value can be calculated as shown below.

8. Root Mean Square (rms)

The rms value can be calculated as shown below. Its numerical value is about 11% higher than that of AA.

9. Flaws: Irregularity which occur at one place or widely varying intervals in surface. It include **cracks, scratches** etc.

X	X ²
3	9
15	225
20	400
33	1089
25	625
18	324
5	25
10	100
15	225
15	225
5	25
11	121
14	196
13	169
27	729
8	64

Total 234 4551

AA = 234/16 = 14.6 micro in.

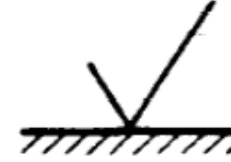
RMS = (4551/16)^{1/2} = 16.9 micro in.



INDICATION OF SURFACE TEXTURE

The **basic symbol** consists of two legs of **unequal** length inclined at approximately **60 degrees** to the line representing the considered surface

The symbol must be represented by thin line



If the **removal of material by machining** is required, a bar is added to the basic symbol,



If the **removal of material is not permitted**, a circle is added to the basic symbol.

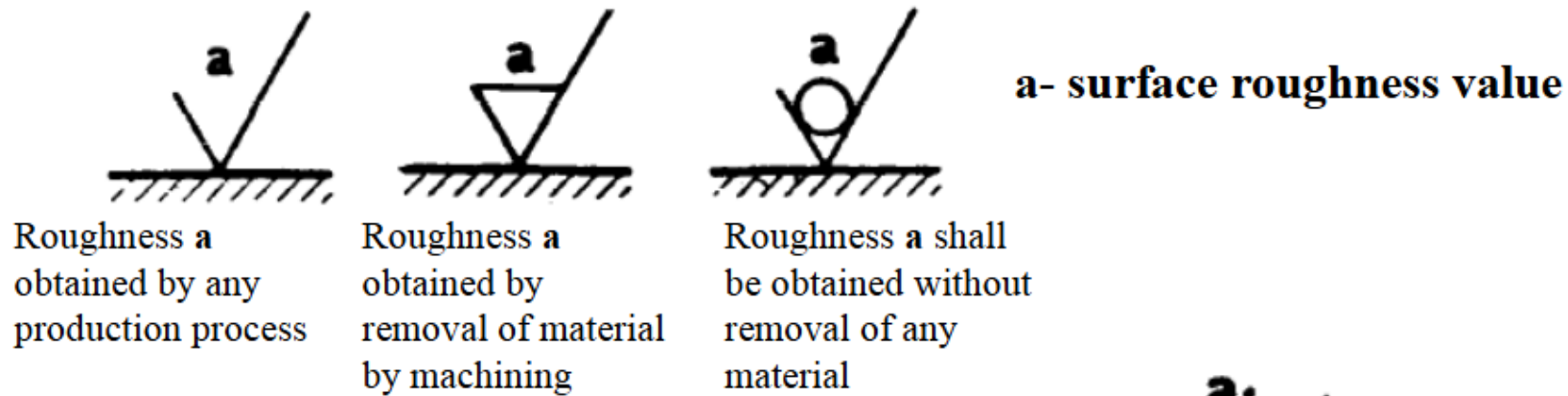


When **special surface characteristics** have to be indicated, a line is added to the longer arm of any of the above symbols,

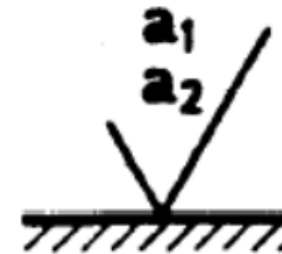


Indication of Surface Roughness

The value or **values** defining the principal criterion of **roughness** are added to the symbols

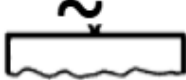
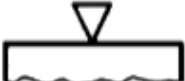


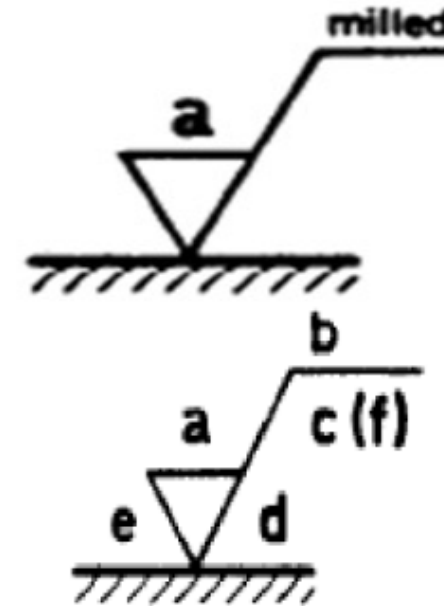
If it is necessary to impose **maximum** and **minimum limits** of the principal criterion of surface roughness, both values shall be shown



maximum limit (a_1) ; minimum limit (a_2).

If it is required that the required **surface texture** be produced by one **particular production method**, this method shall be indicated in plain language on an extension of the longer arm of the symbol

Roughness Value (Microns)	Roughness Grade	Conventional Symbol
50	N12	
25	N11	
12.5	N10	
6.3	N9	
3.2	N8	
1.6	N7	
0.8	N6	
0.4	N5	
0.2	N4	
0.1	N3	



a = Roughness value R_a in micrometres
or
= Roughness grade number N1 to N12.

b = Production method, treatment or coating
 c = Sampling length
 d = Direction of lay
 e = Machining allowance
 f = Other roughness values (in brackets).

Symbol of Lay

Interpretation

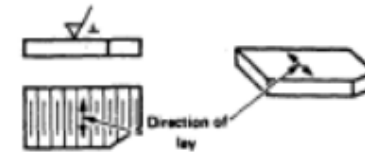
=

Parallel to the plane of projection of the view in which the symbol is used



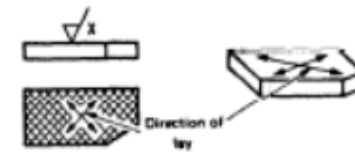
⊥

Perpendicular to the plane of projection of the view in which the symbol is used



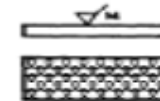
X

Crossed in two slant direction relative to the plane of projection of the view in which the symbol is used



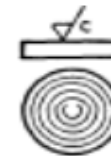
M

Multidirectional



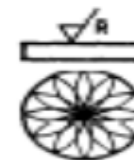
C

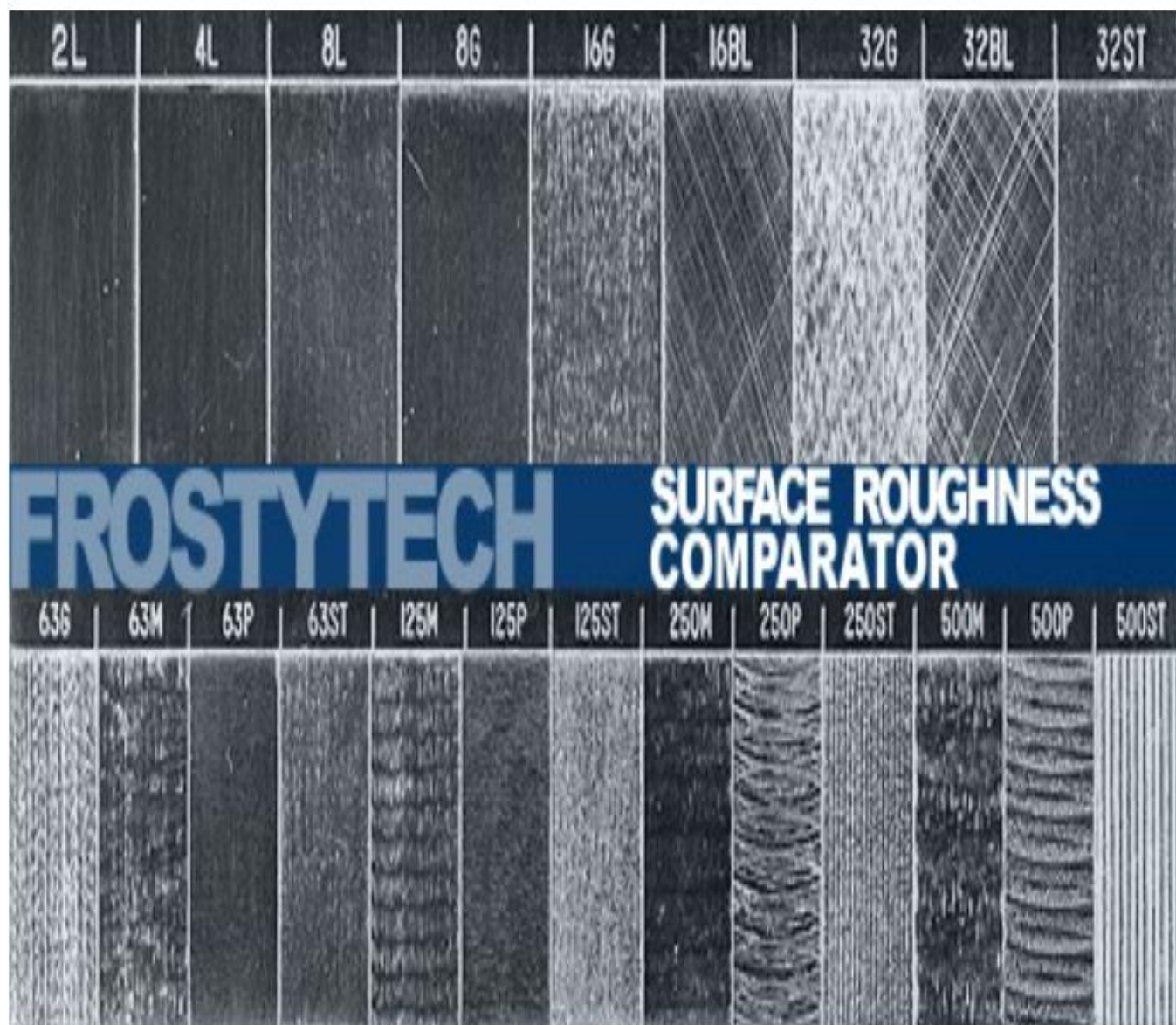
Approximately circular relative to the centre of the surface to which the symbol is applied



R

Approximately radial relative to the centre of the surface to which the symbol is applied





The factors affecting surface roughness

- Type of coolant used
- Cutting parameters such as feed, speed and depth of cut
- Type of machining
- Rigidity of the system consisting of machine tool, fixture, cutting tool and work
- Vibrations
- Material of tool and work piece.

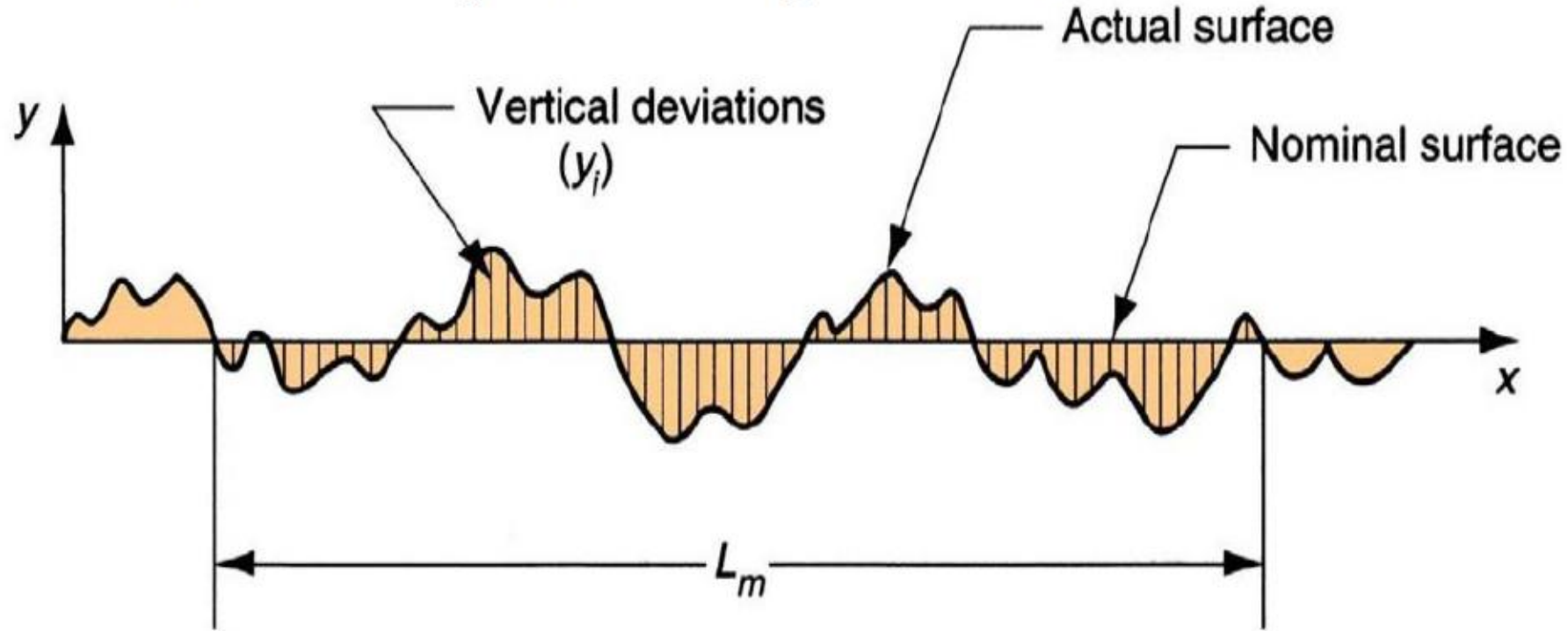
Orders of Geometrical irregularities

Any material being machined by conventional machining process cannot be finished perfectly. The surface generated will have some irregularities and these geometrical irregularities could be classified as follows:

- **First Order:** It includes the irregularities developed due to the **inaccuracies in the machine tool** such as lack of straightness of guide ways, on which tool post is moving.
- **Second Order:** It includes tile irregularities developed due to the **vibrations and rigidity** of machine tools.
- **Third Order:** It includes the irregularities due to the **cutting parameters** such as cutting **speed, feed and depth of cut**.
- **Fourth Order:** It includes the irregularities developed due to the **rupture of the material** during the separation of the chip from the already finished surface of the work piece.

Surface Roughness

Average of vertical deviations from nominal surface over a specified length surface.

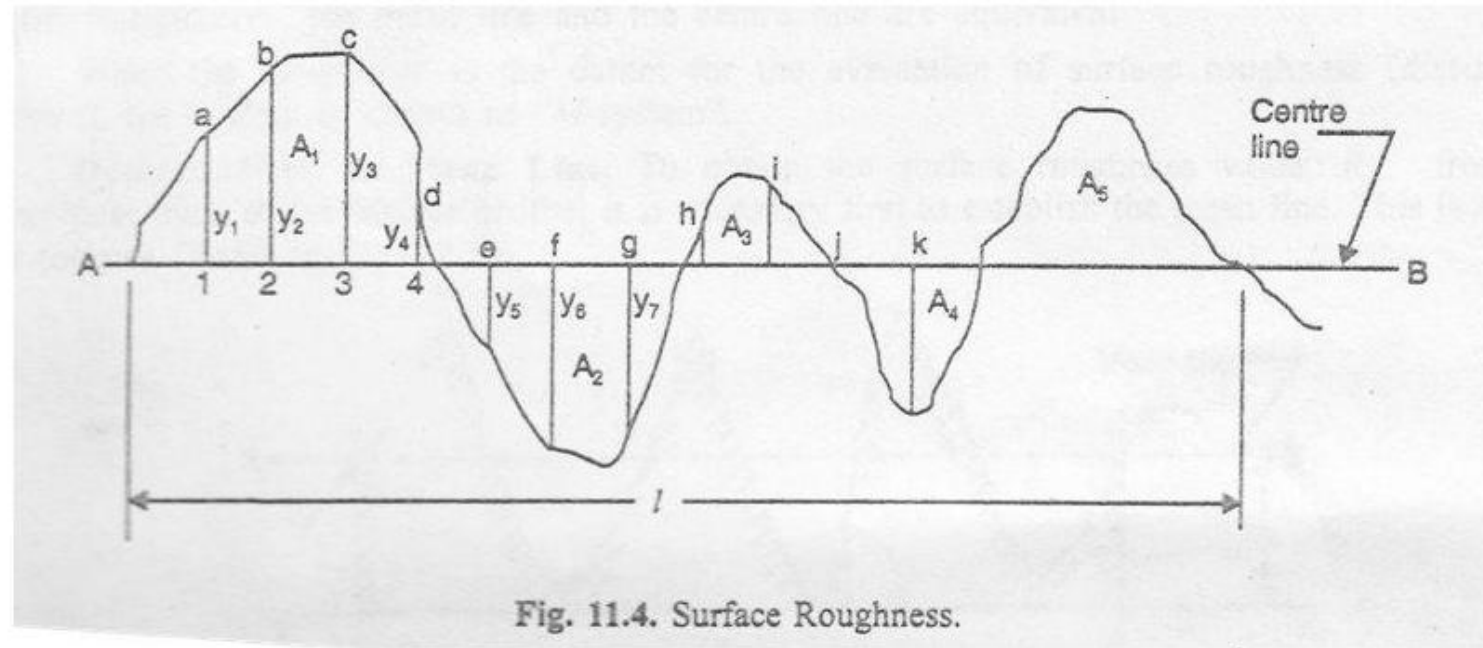


Sampling Length: It is the length of Profile necessary for the evaluation of irregularities to be taken in account. Also called as “Cut-off” length.

It is measured Parallel to direction of profile.

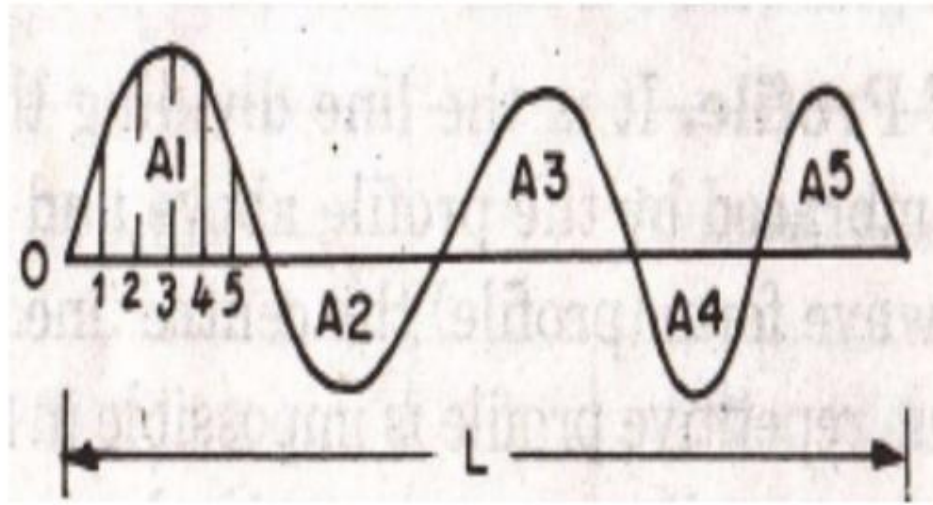
Mean line of the Profile: It is the line that divides the effective profile such that, within sampling length the sum of squares of vertical ordinates ($y_1, y_2, \dots$) between effective profile points & mean line is minimum or Zero.

Center line of the Profile: It is the line that divides the effective profile such that, the area contained by the profile above & below the line are equal.



Analysis of Surface Traces(surface finish)

1. Centre line Average (C.L.A. Method)



- Surface roughness is measured as the **average deviation from nominal surface**.

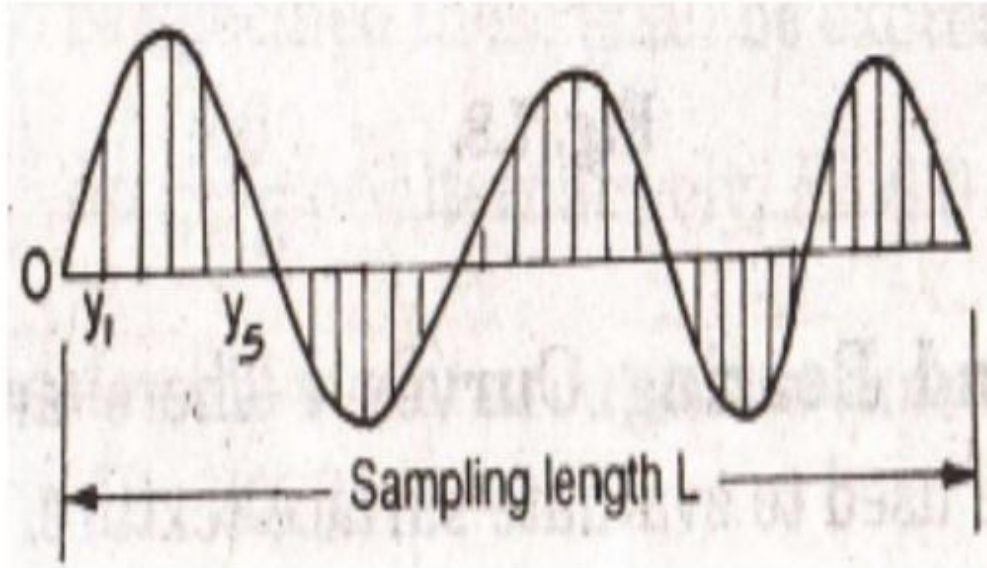
- It is defined as **average value of the ordinates** from mean line, **regardless** of the arithmetic **sign** of ordinates.

$$C.L.A. Value = \frac{h_1 + h_2 + h_3 + \dots + h_n}{n} \quad (1)$$

Calⁿ of CLA Value by equation 2 is carried out by **Planimeter**.

$$C.L.A. Value = \frac{A_1 + A_2 + A_3 + \dots + A_n}{L} = \frac{\sum A}{L} \quad (2)$$

2. Root Mean Square (R.M.S. Method)



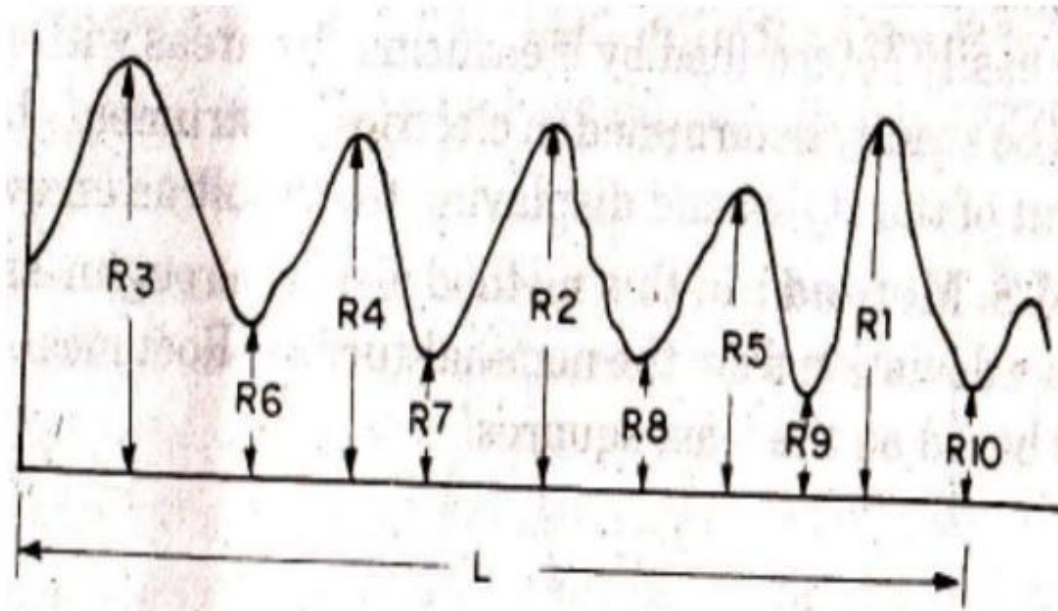
$$R.M.S.Avg. = \sqrt{\frac{y_1^2 + y_2^2 + \dots + y_n^2}{n}}$$

$$y_{rms} = \sqrt{\left[\int_0^L y^2 dl \right]}$$

- Surface roughness is measured as the **average deviation from nominal surface**.

- It is defined as **square root of arithmetic mean of values of square of ordinates** of the surfaces measured from mean line.

3. Ten Point Height Method



In this method avg. difference between five height peaks & five lowest valleys of surface texture within sampling length measured from a line parallel to mean line & not crossing the profile.

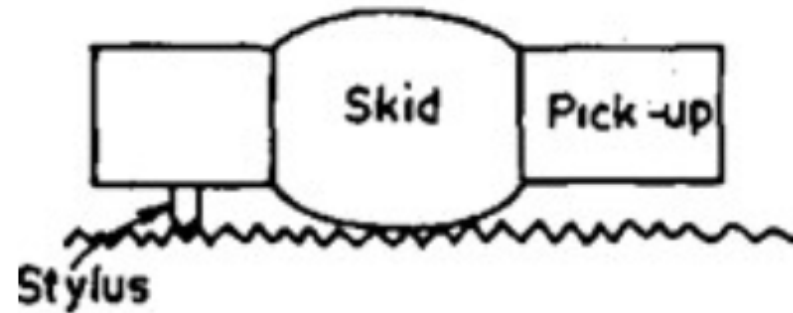
It is simple method to measure total depth of surface irregularities, but it does not give sufficient information about surface, as no account is taken of frequency of the irregularities.

If R = Ten point height of irregularities then,

$$R = \frac{1}{5} [(R_1 + R_2 + R_3 + R_4 + R_5) - (R_6 + R_7 + R_8 + R_9 + R_{10})]$$

Measurement of Surface finish

1) Stylus Probe Instrument (Contact type)



- These methods enable to determine a numerical value of the surface finish of any surface.
- Skid or shoe which is drawn slowly over the surface either by hand or by motor drive. The skid when moved over the surface follows its general contours and provides a datum for the measurements.

- A stylus or probe which moves over the surface with the skid. The stylus should be cone shaped with a spherical tip. This records the micro-geometrical form of the surface. It moves vertically up and down relative to skid movement due to roughness of the surface.
- As the stylus tracks the surface peaks and valleys, its vertical motion is converted to a time varying electrical signal that represent surface profile.
- Generally it is desired that if the skid is moving up then the stylus must also be moving up.

Evaluation of Surface Integrity

- ***Surface texture.*** Surface roughness, designation of lay, and other measures provide superficial data on surface integrity. This type of testing is relatively simple to perform and is always included in the evaluation of surface integrity.
- ***Visual examination.*** Visual examination can reveal various surface flaws such as cracks, craters, laps, and seams. This type of assessment is often augmented by fluorescent and photographic techniques.
- ***Microstructural examination.*** This involves standard metallographic techniques for preparing cross sections and obtaining photomicrographs for examination of microstructure in the surface layers compared with the substrate.
- ***Microhardness profile.*** Hardness differences near the surface can be detected using microhardness measurement techniques such as Knoop and Vickers (Section 3.2.1). The part is sectioned, and hardness is plotted against distance below the surface to obtain a hardness profile of the cross section.
- ***Residual stress profile.*** X-ray diffraction techniques can be employed to measure residual stresses in the surface layers of a part.

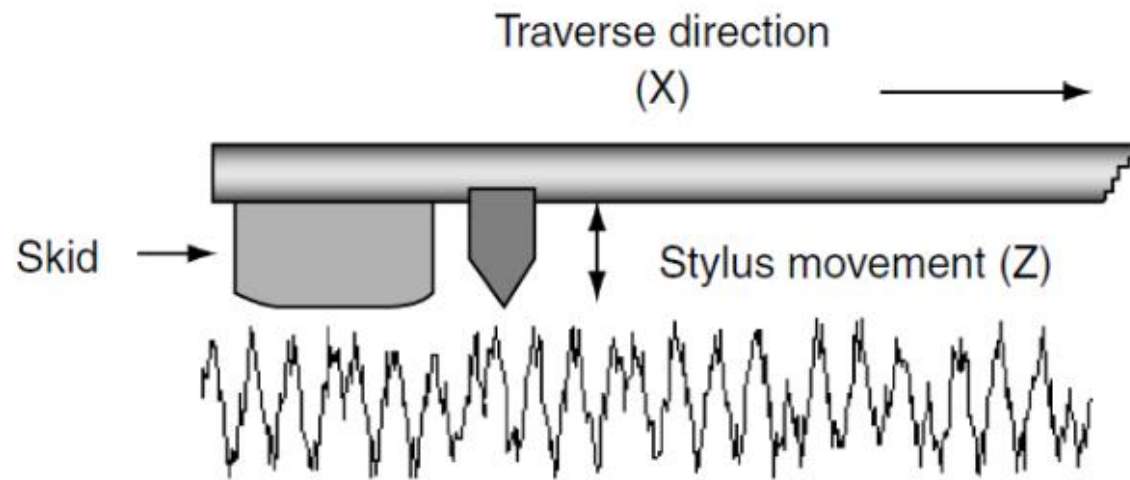
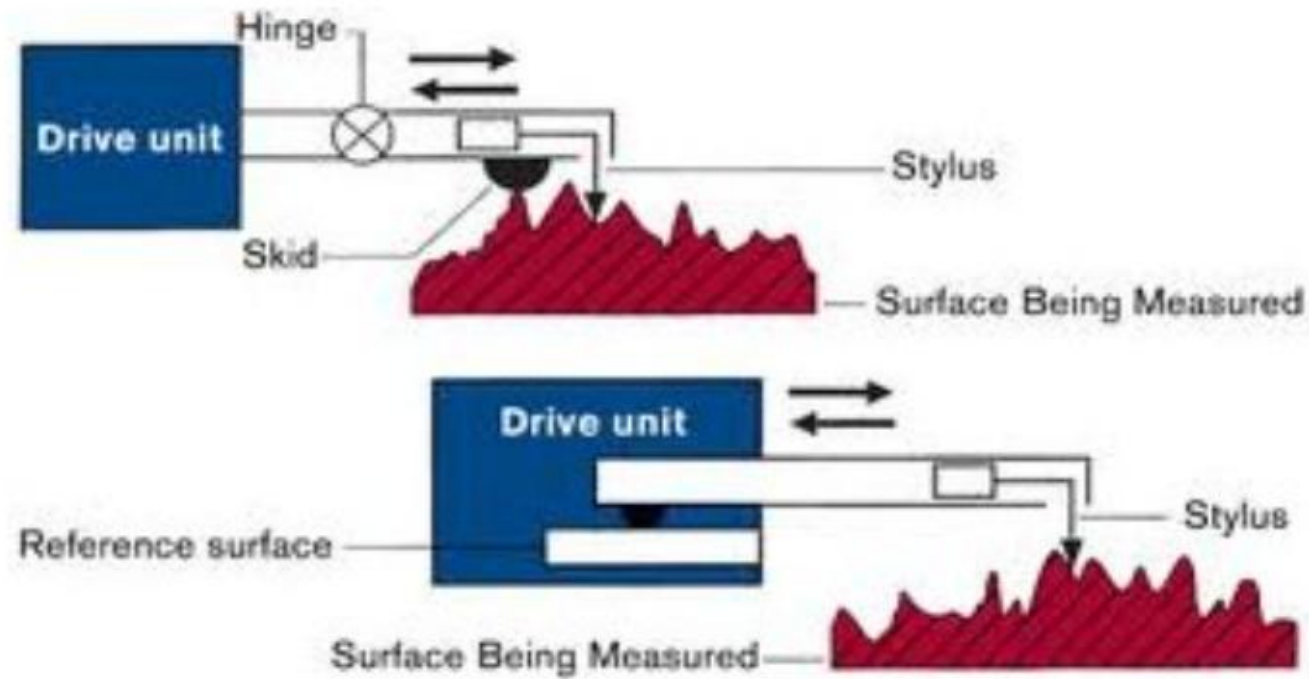
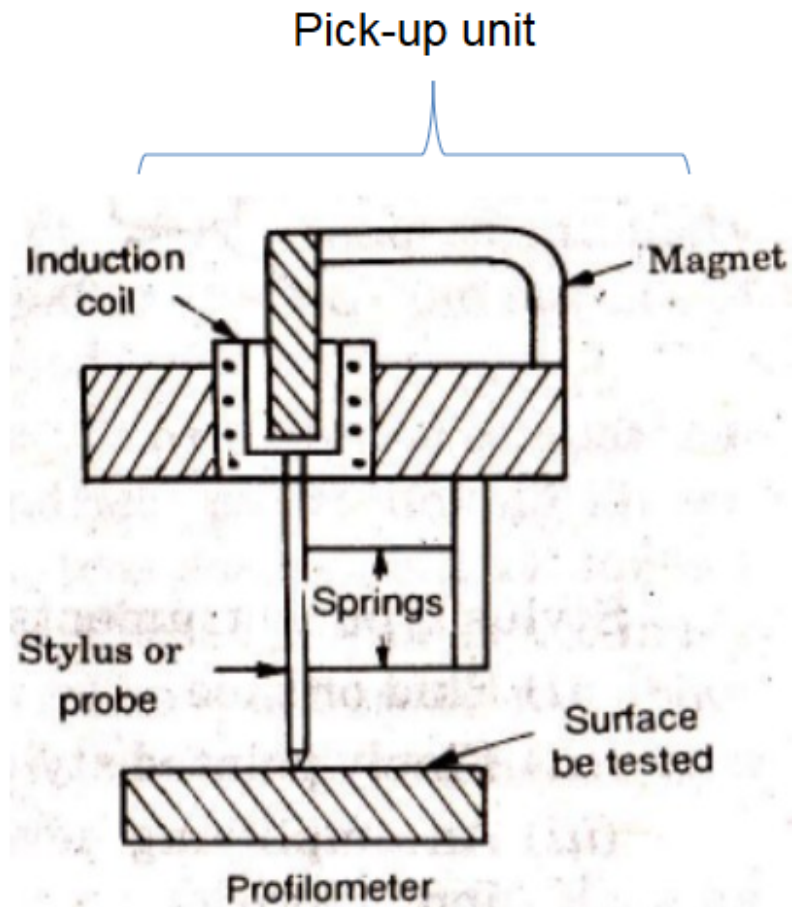


Fig. 10.7 *Skid*



2) Profilometer (Contact type)



- It is indicating & recording instrument used to measure roughness in microns.

- It consist of two units Tracer & amplifier.

- Tracer is finely pointed stylus mounted in pick-up unit which consist of induction coil located in field of Permanent magnet.

- When tracer(stylus) is moved across the surface to be tested, it is displaced vertically up & down due to surface irregularities.

- This causes induction coil to move in the field of Permanent magnet & induces the voltage.

- The induced voltage is amplified & recorded.

Coordinate Measuring Machine

A classic, bridge-style coordinate Measuring Machine (CMM) accurately measures the geometry of an object along X, Y and Z axes using a touch-trigger, scanning or vision probe to take a series of precise points on the surface of an object.



Thank You...